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Editorial Board
Journal of Statistical Software

Dear Editors,

We are pleased to submit our manuscript titled "**StickForStats: A Statistical Analysis Platform with Automatic Assumption Validation**" for consideration as a software paper in the Journal of Statistical Software.

Summary of the Work

Statistical assumption violations remain a persistent problem in published research. While assumption testing tools have been available in statistical software for decades, their optional nature means researchers often skip validation. Our platform addresses this through the Guardian system, which automatically validates assumptions before every statistical test without requiring user action.

The key contributions of this work are:

- **Guardian System:** An automatic assumption validation layer implementing 15+ validators covering normality, variance homogeneity, independence, outliers, linearity, homoscedasticity, sphericity, multicollinearity, and more.
- **AI Statistical Advisor:** A natural language interface providing guidance for test selection, result interpretation, and generating JARS-compliant methods sections automatically.
- **Paper Parser with Statistical Debugger:** A manuscript analysis tool that identifies statistical reporting errors, assesses compliance with JARS-Quant guidelines, and provides a Statistical Quality Score prototype for pre-submission quality assessment.
- **High-Precision Computing:** Optional 50-decimal-place precision using mpmath for exact p-value calculations when standard floating-point arithmetic is insufficient.
- **Validated Implementation:** Computational accuracy verified against SciPy, R, and statsmodels to 14+ decimal places across all statistical tests.
- **Real Data Validation:** Case studies with Fisher's Iris, UCI Wine Quality, and classic R datasets demonstrating practical value.

Relevance to JSS

StickForStats is an open-source platform built with Python (Django backend, React frontend). The software is available at `https://github.com/visvikbharti/stickforstats_new` under an open-source license. We believe this work aligns well with JSS's mission of publishing statistical software that advances reproducible research.

Replication Materials

We have included Python scripts that reproduce all numerical results presented in the paper. These scripts validate our computations against SciPy and demonstrate Guardian's assumption checking on real datasets.

Confirmations

- This manuscript has not been published elsewhere and is not under consideration by another journal.
- All authors have approved the manuscript and agree to its submission.
- The software and all results are fully reproducible.

We thank you for considering our submission and look forward to your response.

Sincerely,

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